Yanisse FERHAOUI

Software Engineer — 3D — Computer Vision

Phone : +33 7 82 79 99 67 yanisseferhaoui@gmail.com
linkedin.com/in/yanisse-ferhaoui
github.com/YanisseFrh
yanisse-ferhaoui.fr

Technical Skills

Core & 3D Languages C++, C, Python, GLSL, Qt/QML, Java
Web & Fullstack JavaScript, React.js, PHP/Symfony, HTML/CSS
Specialized Libraries OpenGL, OpenCV, SDL2, VTK/ITK, PyTorch, TensorFlow
Engineering & DevOps Git, GitLab CI/CD, Docker, CMake, Linux, GoogleTest
Languages Français (native), Anglais (professional)

Education

Master's in Computer Science – Image, Development and 3D Technologies — 2023 – 2025
Claude Bernard University Lyon 1, Villeurbanne

M2 : GPU programming, Virtual Reality, Image processing, Computational geometry.
Tronc commun M1 : Software engineering & project management, Artificial Intelligence, Computational complexity, Design Patterns.

Bachelor's in Computer Science — 2020 – 2023
Claude Bernard University Lyon 1, Villeurbanne

Comprehensive CS foundation : Operating systems, IP/TCP networks, SQL databases, Computer architecture, Web development, Advanced algorithms.

Professional Experience

Institut Pascal – Université Clermont Auvergne — Feb. 2025 – Jul. 2025
R&D Intern : Artificial Intelligence & Software Integration — Le Puy-en-Velay

- Architectural integration of LLMs into the open-source medical software 3D Slicer.
- Training and fine-tuning of deep learning models on a multi-GPU compute cluster.
- Development of an extension for user assistance.

AMS Association Mantes Solidarité — May 2023 – Jun. 2023
Fullstack PHP/Symfony Developer — Mantes-La-Ville

- Integration of a payment form and a donation management interface for administrators.

Selected Projects

Mesh Viewer | C++, Qt, OpenGL, CMake — Sep. 2025 – Today

- Design of a high-performance mesh viewer supporting standard formats (.obj, .off).
- Implementation of the graphics pipeline and texture management.
- Setup of a GitLab CI/CD pipeline for automated builds, tests and deployment.

SlicerGPT | Python, Transformers, Qt — Mar. 2025 – Sep. 2025

- 3D Slicer extension integrating a local chatbot.
- Provides context-aware assistance using the scene and the official documentation.

Computational Geometry | C++, Data Structures — Sep. 2024 – Nov. 2024

- Implementation of Lawson's algorithm and mesh manipulation.
- Optimization of data structures for fast processing of point clouds.

Embedded Robotics & Vision | C++, Linux EV3Dev, OpenCV — Feb. 2024 – May 2024

- C++ development on an embedded system for an autonomous robot.
- Real-time image processing for object detection and multi-camera data fusion.
- Management of inter-process communication and motor control.